

2023 Ch H1 Q6

Section: Chemistry in Society

Topic: Equilibria

Question Summary

The diagram shows the concentration of a reactant against time during a reversible chemical reaction. Which diagram best represents the effect of adding a catalyst?

Worked Solution

A catalyst speeds up both the forward and reverse reactions equally. This means equilibrium is reached more quickly, but the position of equilibrium (the final concentrations) remains unchanged.

Therefore, the correct diagram is the one where the equilibrium concentration is the same, but the curve is steeper and levels off sooner.

Final Answer

C — Equilibrium reached faster, same final concentration

Revision Tips

- A catalyst lowers activation energy, increasing the rate of both forward and reverse reactions.
- Catalysts do not affect the position of equilibrium or ΔH .
- Graphically: equilibrium curve is steeper, same final concentration.